

1.1 Types and Tokens

Expressions. Ideas, reasoning, information are all communicated by means of language. The objects of language by which we communicate them are called *expressions*.

Notion 1.1 (Expression). An **expression** is a reproducible pattern of symbols that can be written or printed or displayed or uttered.

Letters, numerals (names for numbers), digits (one-character numerals), words, sentences, paragraphs, for instance, count as expressions, and so do made-up symbols or any other combination with no established meaning. In what follows, we will focus on printable or displayable expressions, rather than on utterable ones.

Type, token, and occurrence. A distinction may be made between a concrete appearance of an expression and its abstract shape, that is, between an *expression token* and an *expression type*.

Notion 1.2 (Token). An **expression token** is a physical instance of an expression, e.g., a printing of the expression on a page.

Notion 1.3 (Type). An **expression type** is the abstract entity that expression tokens instantiate.

For instance, each of the following three lines display distinct tokens of the same expression type:

bacaba
bacaba
bacaba

Unless otherwise specified, when we speak of an expression, we will refer to its type rather than to a token.

A common but highly misleading conflation is that between an expression token and an *expression occurrence*.

Historical note: Types and Tokens in Typography

In typography, *types* are the basic moulds from which text is imprinted (see figure). By contrast, a *token* is a particular, concrete mark printed from that mould.



Notion 1.4 (Occurrence). An **expression occurrence** is the appearance of an expression type within an expression token or within an expression type.

For instance, in the expression 'bacaba' (whether considered as this specific token or as the type that this token instantiates), there is one occurrence of the letter (type) 'c', there are two occurrences of the letter 'b', and there are three occurrences of the letter 'a'. In other words, the letters 'a', 'b', and 'c' occur one, two, and three times, respectively, in the word 'bacaba'.

Task 1.1 (Tokens and types). Consider the sentence:

Aristotle was a philosopher who lived in Ancient Greece.

Answer the following questions:

1. Which letter types occur in the sentence?
2. How many letter types does it contain?
3. How many tokens correspond to each of letter type?
4. How many letter tokens does it contain in total?

Treat each letter design as a distinct expression type – so that, e.g., the expressions 'a' and 'A' would count as different types.

To answer **task 1.1.1**, we treat uppercase and lowercase letters as distinct character types. Thus, the distinct letter types in our sentence are the expressions 'A', 'G', 'a', 'c', 'd', 'e', 'h', 'i', 'l', 'n', 'o', 'p', 'r', 's', 't', 'v', and 'w'. That is, we have 17 letter types, which answers **task 1.1.2**.

To answer **task 1.1.4**, we count the times each letter type identified occurs in our sentence (see table on the next page). Finally, to answer **task 1.1.4**, we add up these values, giving a total of 47.

Well-formed and ill-formed expressions. Not all expressions can perform a communicative function within a language. Each language – whether natural or artificial – has certain conventions,

Solution to task 1.1.

1. The distinct letter types in the sentence are: 'A', 'G', 'a', 'c', 'd', 'e', 'h', 'i', 'l', 'n', 'o', 'p', 'r', 's', 't', 'v', 'w'.
2. There are 17 letter types in total: 2 uppercase letters and 15 lowercase letters.
3. The number of tokens corresponding to each letter type is indicated in the table.
4. The total number of letter tokens equals 47.

Types	Tokens
e	7
i	5
o	4
h, l, n, r, t	3
A, a, c, p, s, w	2
G, d, v	1

Historical note: Type-Token Distinction in Logic

Charles S. Peirce (1839–1914)

The distinction between token and type was first introduced by Charles Sanders Peirce in a paper entitled 'Prolegomena to an apology for pragmatism' (1906, pp. 505–6):

There will ordinarily be about twenty *thes* on a page, and of course they count as twenty words. In another sense of the word "word", however, there is but one word "the" in the English language; and it is impossible that this word should lie visibly on a page. ... Such a definitely significant Form, I propose to term a *Type*. A Single event which happens once and whose identity is limited to that one happening ... I will venture to call a *Token*.

such as rules of syntax, that determine whether a given expression can fulfil that function. A symbol fulfilling such conventions counts as a *well-formed expression* of that language, while one which does not is an *ill-formed expression*.

Names, phrases, syntagms, sentences, and the like all count as well-formed expressions. Moreover, some symbols that lack meaning in English – like, e.g., the phrase 'the colour of my favourite number' – may also count as well-formed expressions in English.

However, many expressions – like, e.g., as 'favourite my the the my', 'AKARaaaA', and '\$)&(((' – do not qualify as well-formed expressions in English. Indeed most expressions are ill-formed rather than well formed with respect to any language.

Now, the distinction between *well-formed* and *ill-formed expressions* is quite specific to particular languages rather than to language in general. For this reason, we shall not explore it in depth for now.

1.2 Meaning and Quotation

Meaningful expressions. So far, we have spoken about expressions as such, which may or may not be well-formed. However, not every expression – even a well-formed one – is meaningful.

Consider, for example, the following expression:

the colour of the natural numbers.

Although grammatically well-formed in English, this expression does not refer to anything, since neither numbers nor sets of numbers have colours.

There are many different theories about what it means for an expression to have a meaning. Some theories connect meaning with mental ideas, others with use in language, and still others with truth-conditions.

We cannot settle these questions here. Instead, we will focus on a simpler idea: *when an expression is meaningful, it stands for something*. The thing a meaningful expression stands for is its *referent*.

Notion 1.5 (Meaningful expression). A **meaningful expression** stands for something, which is its **referent**.

The referent of an expression may be a physical object, such as a planet or a person; it may be a mathematical object, such as a number or a set. Some have even suggested that expressions may refer to fictional entities, such as a character in a story, or even impossible entities, such as a round square.

In natural language, the referent of an expression may vary because a single expression can have various meanings. For instance, the expression 'bank' may refer to a financial institution, or to the side of a river. Likewise, the expression 'bat' may refer to a flying mammal, or to a piece of equipment used in certain sports.

In ordinary language, ambiguity of this kind is very common, and usually the intended referent is determined by the larger expression within which the original expression occurs.

In contrast, the formal languages that we will study are designed to avoid such ambiguity. Each meaningful expression in a formal language is introduced with a fixed meaning, so that it always stands for one referent, regardless of the context, and not for different ones, which may vary according to the context.

For this reason, in what follows we will speak as if every meaningful expression has exactly one referent.

Task 1.2. Determine the referent of each of the expressions:

1. Earth's natural satellite
2. 2^3
3. Sherlock Holmes' flatmate
4. the largest prime number
5. the freezing point of water
6. the present king of France
7. The Beatles' first album
8. the number of letters in the word 'logic'

If an expression has no referent, state so.

Historical note: The Theory of Supposition

William of Ockham
(ca. 1287–1347)

It is difficult to attribute the introduction of the notions of *use*, *mention*, *standing for*, and *reference*, as we understand them today, to a particular source. Rather, they appear to have emerged gradually, with antecedents traceable to Aristotle.

A milestone in their development was reached in William of Ockham's *theory of supposition*, formulated by in his *Summa Logicae* (*Sum of Logic*, [ca. 1323](#)). The following is an extract:

Personal supposition, in general, occurs when a term stands for (*supponit pro*^a) the thing it signifies...

Material supposition is when a term does not stand signifi-
catively, but instead stands either for something spoken
or written. This is clear in 'man is a name', where the term
'man' stands for itself, and yet does not signify itself. (I.64)

^a More accurate translations prefer 'supposits for'.

Use and mention. We have determined to speak as if every meaningful expression had exactly one referent. There is nevertheless one notable exception we want to make to this rule.

Sometimes, we wish to speak about expressions themselves rather than about their referents. For example, we might wish to employ the word 'Berlin' not to speak about its referent, the city Berlin, but about the word 'Berlin', which contains six letters.

In such cases, the word 'Berlin' is not being *used* to express its referent, but is instead *mentioned* as a linguistic object.

Notion 1.6 (Use). A meaningful expression is **used** within another when the former occurs within the latter standing for its referent.

Notion 1.7 (Mention). Something is **mentioned** within a meaningful expression when an expression occurs within the latter standing for the former.

Note that, the notion of *mention* applies to any object whatsoever, including, e.g., objects in the real world and expressions. Instead, the notion of *use* applies only to meaningful expressions. Thus, we cannot use an object in this sense, as objects have no referent – though they can be the referent of an expression.

To exemplify, consider the following sentences:

Alice is a great musician. (7)

Alice has five letters. (8)

Solution to task 1.2.

1. the Moon
2. 8
3. John H. Watson or
no referent
4. no referent
5. 0° Celsius
6. no referent
7. the album *Please
Please Me*, of 1963
8. 5

In sentence (7), the name 'Alice' is being used to talk about Alice, the person. In sentence (8), instead, the name 'Alice' is being mentioned: we are talking about the word itself rather than using it to speak about its referent, (the person) Alice. Alice herself is only mentioned in (7). In fact, she cannot be used in the sense of **notion 1.7**, because Alice is not an expression – her name is.

Moreover, we used the symbols enclosed in parentheses in the previous paragraph – namely, '(7)' and '(8)', with parentheses included – to mention the expressions that appear to the left of their first occurrence in this section – namely, the sentences 'Alice is a great musician' and 'Alice has five letters'.

Also note that the enumerated expressions in **task 1.2** are being mentioned (through themselves), while the expressions in its solution are being used to express their referent.

Task 1.3 (Use and mention). For the sentence below, determine whether the underlined expression is being used or mentioned at that particular occurrence.

1. Socrates is mortal.
2. Socrates has three syllabi.
3. Dogs is the plural of dog.
4. Dogs bark.
5. Athens begins with the letter A.
6. Athens is the name of a Greek city.
7. Aristotle is not short.
8. The name of Aristotle is not short.

In **sentence 1.3.1**, the name 'Socrates' refers to the philosopher, thus being used. In contrast, in **sentence 1.3.2**, the name 'Socrates' is not employed to speak about the person but about the word itself, and it is thus being mentioned through itself.

A similar pattern appears in the next pair. In **sentence 1.3.4**, the word 'Dogs' mentions a kind of animal, thus being used. But in **sentence 1.3.3**, it refers to a word in English, thus being mentioned.

The word 'Athens' is mentioned rather than used in both **sentences 1.3.5** and **sentence 1.3.6**. Note, however, that although Athens, the city, is mentioned in **sentence 1.3.6**, we do not do so via the name 'Athens', but via the phrase 'a Greek city'. Hence, we are not using the name 'Athens' in **sentence 1.3.6**.

In **sentence 1.3.7**, the name 'Aristotle' has an ambiguous reference. The sentence may be interpreted either as using the name 'Aristotle' to say that the philosopher was small in size, or to say of that the

Historical note: Use and mention

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX–XXXX)

philosopher's name is brief in length. Hence, depending on the interpretation the name 'Aristotle' may be used or mentioned.

Now, in **sentence 1.3.8**, the phrase 'the name of Aristotle' refers to an expression, i.e., the name 'Aristotle'. However, the occurrence of the name 'Aristotle' within it refers to the philosopher. Hence, we use the name 'Aristotle' to mention Aristotle, but also mention the name 'Aristotle' by using the phrase 'the name of Aristotle'.

Quotation. The underlined expressions in **task 1.3** sometimes denote their referents and are sometimes used to denote themselves, depending on the sentence. More generally, expressions in natural language can be used either to convey their usual meanings or to refer to themselves as linguistic expressions.

Unfortunately, as **sentence 1.3.7** illustrates, it is not always easy to determine which meaning is intended. We have avoided those ambiguities by *using* expressions such as 'the word', 'the name', and 'the phrase', followed by the *mentioned* expression, enclosed in quote marks. Alternatively, we have introduced the mentioned expressions by means of a colon, placing it on a new, centred line.

From now on, we will explicitly adopt the *quote marks* notation.

Solution to task 1.3.

Used: 1, 4, 6, 8.

Mentioned: 2, 3, 5.

Ambiguous: 7.

Notation 1.8 (Quotation). An expression can be mentioned via another expression formed by: first, a left quote mark, next, the original expression, and, finally, a right quote mark.

For instance, to mention the first letter of the alphabet we use 'a', and to mention this latter expression we use "a", and so forth.

To explain this notation, I will use a visual trick: the sub-expression inside the quote marks of some quoted expressions will be coloured in order to show exactly what is being mentioned.

'Queen are legendary' mentions Queen and uses 'Queen'. "Queen' has five letters' mentions 'Queen' and uses "Queen". To mention Queen we use 'Queen', to mention 'Queen' we use "Queen", and to mention "Queen" we use "'Queen'". 'Queen' has five letters. "Queen" has seven characters: a left quote mark, five letters, and a right quote mark. Finally, note that 'Queen' names and mentions Queen, a legendary British rock band.

If this feels tricky, you are not alone, so take your time to grasp it. And even if you feel confident, pause and reflect again! Quotation is one most difficult things to grasp in logic, so patience pays off.

Task 1.4. Rewrite the sentences of **task 1.3** using quote marks. (Pay attention to non-underlined expressions too.)

The only sentences to modify are those mentioning expressions, that is: **sentences 1.3.2, 1.3.3, 1.3.5, 1.3.6**. All their underlined expressions should be placed between one pair of quote marks.

Additionally, **sentence 1.3.3** should enclose 'dog' in quote marks, since 'dogs' is not said to be the plural of some particular dog. Similarly, **sentence 1.3.5** should enclose 'A' in quote marks, as we are referring to the letter itself, and not to any referent that might be ascribed to 'A' in this or any other context.

Solution to task 1.4.

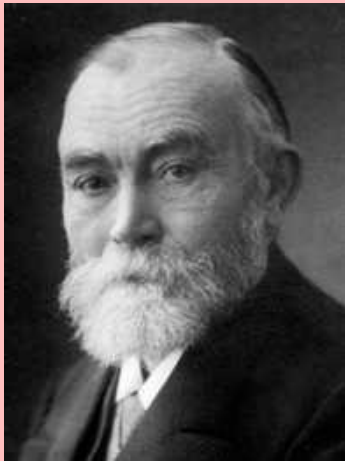
1. Socrates is mortal.
2. 'Socrates' has three syllabi.
3. 'Dogs' is the plural of 'dog'.
4. Dogs bark.
5. 'Athens' begins with the letter 'A'.
6. 'Athens' is the name of a Greek city.
7. Aristotle is not short. 'Aristotle' is not short.
8. The name of Aristotle is not short.

Quote marks make it easy to refer to the empty expression.

Notion 1.9 (Empty expression). The **empty expression** is the expression which contains no symbols.

Notation 1.10 (Empty expression). We mention the empty expression with **" "**. Thus, **" "** (i.e., the nothing between the quote marks) is the empty expression.

The empty expression is not the same as the blank space ' '.

Historical note: Quote marks

Gottlob Frege (1848–1925)

The modern use of quote marks in logic appears to have been first clearly articulated by Gottlob Frege in his paper ‘Über Sinn und Bedeutung’ (On sense and reference, 1892, p. 23):

When using words in the ordinary way, we wish to speak about their reference [*Bedeutung*]. But it may also happen that we wish to speak about the words themselves... We then have signs of signs. In writing, in such a case, we enclose the word-forms in quote marks. A word-form standing within quote marks must therefore not be taken in its ordinary reference. If we wish to speak of the sense [*Sinn*] of an expression ‘A’, we can do so simply by using the phrase ‘the sense of the expression ‘A’.

The name of the name of the name. Sometimes, we do not want to speak about the name of Socrates – that is, about ‘Socrates’ – but about the name of ‘Socrates’ – that is, about the name of the name of Socrates. Moreover, we may want to speak about the name of the name of the name of Socrates, or about the name of the name of the name of the name of Socrates. and so forth.

In such cases, remember that we can just add around our expression another pair of quote marks, as in “Socrates”, and another pair of quote marks, as in ““Socrates””, and so forth.

Task 1.5. Decide whether the following are true or false:

1. Aristotle is a philosopher.
2. Aristotle has nine characters.
3. Aristotle’s name has nine characters.
4. ‘Aristotle’ has nine characters.
5. ‘Aristotle’ is a philosopher.
6. “Aristotle” has nine characters.
7. “Aristotle” has eleven characters.
8. ““Aristotle”” is the name of the name of “Aristotle”.

Under the convention just proposed, it is clear that sentences 1.5.1, 1.5.3, 1.5.4, and 1.5.7 are true. In fact, note that sentences 1.5.3 and 1.5.4 mean the same, as Aristotle’s name is ‘Aristotle’.

Sentence 1.5.6 is clearly false. The expression referred to here is “Aristotle” and not ‘Aristotle’, since removing one pair of quote marks around “Aristotle” results in the former expression rather than in the latter.

Sentence 1.5.8 is also true, since, in our convention, the name of x is ' x '. Hence, the name of the name of x would be " x ". Thus, if we replace ' x ' with "'Aristotle'", we obtain what **sentence 1.5.8** says.

Finally, **sentences 1.5.2** and **1.5.5** may be regarded as false or nonsensical, depending on the account of truth we assume.

1.3 Sentences and Propositions

Sentences. We have used and mentioned several sentences so far, but we have not yet explained what is a sentence.

Notion 1.11 (Sentence). A **sentence** is an expression that is complete in the sense that it expresses a statement.

The following expressions are all examples of sentences:

'The door is closed.'	($\top$)
'Is the door closed?'	($\oslash$)
'Close the door!'	($\perp$)

Each of these expressions is complete in the sense that it suffices (in appropriate contexts) to perform a full communicative act.

Sentence ($\top$) is *declarative*, as it declares that the door is closed, which may be true or false. Sentence ($\oslash$) is *interrogative*, as it poses the question whether the door is closed, which has a specific true answer. Finally, sentence ($\perp$) is *imperative*, as it commands that the door be closed, which may or may not be carried out.

In each case, nothing further needs to be added in order to understand what is being said, asked, or demanded. In this sense, we might say that a sentence is, in general, an expression conveying a complete piece of information.

Sentences and propositions. Our first focus will be on declarative sentences. However, we will not consider any sentence in declarative form, but only *propositional sentences*, that is sentences expressing something that can be true or false or *propositions*.

Notion 1.12 (Proposition). A **proposition** is something that can be true or false.

Notion 1.13 (Propositional sentence). A **propositional sentence** is a sentence that stands for a proposition.

Remember that two expressions formed by different symbols are different even if they share the same meaning. Thus, two sentences may be different even if they express the same proposition. Take,

Solution to task 1.5.

True: 1, 3, 4, 7, 8.

False: 6.

Nonsensical or false: 2, 5.

Historical note: Judgement and the movements of the soul

A. Arnauld (1612–1694)

The classification of sentence types into declarative, interrogative, and imperative has a precursor in the Port-Royal's *Grammaire générale et raisonnée* (*General and Rational Grammar*, 1660, p. 19), by Antoine Arnauld and Claude Lancelot:

And thus the most important distinction concerning what takes place in our mind is that we may consider there the object of our thought, and the form or manner of our thought, of which the principal one is judgment. But we must also include conjunctions, disjunctions, and other similar operations of our mind, as well as all the other movements of our soul, such as desires, command, interrogation, etc.

for instance, the following three different descriptive sentences which express one and the same proposition:

‘Hamlet was written by William Shakespeare.’

‘William Shakespeare is the author of Hamlet.’

‘Hamlet is a work by William Shakespeare.’

Propositions may be thought as expressing states of affairs. However, those states of affairs need not concern particular facts about the world we live in. They may also concern, e.g., general laws about it or even mathematical theses about any possible world.*

In any case, note that it is necessary for a sentence in declarative form to be meaningful in order to express a proposition.

*: Some even argue that there are propositions about fictional objects, but we will leave those aside.

Task 1.6 (Propositional sentences). Determine which of the following sentences are propositional.

1. The Moon orbits the Earth.
2. Is it going to rain today?
3. All squares have four sides.
4. Please pass the salt.
5. The current king of France is bald.
6. If it has rained, then the floor is wet.
7. Munich is the capital of Germany.
8. This sentence has five words.

Sentences 1.6.1, 1.6.3, and 1.6.7 are straightforward cases: they are propositional sentences stating facts about the world or mathematics, and their truth-values can be clearly assessed.

Sentence 1.6.6 is more interesting, as it asserts that the floor is wet, but conditionally upon it having rained. Hence, it is a propositional sentence, although it has a conditional structure.

In contrast, **sentences 1.6.2** and **1.6.4**. **Sentence 1.6.2** is interrogative, as it asks a question. **Sentence 1.6.4** is imperative, instructing the reader to perform an action.

Finally, **sentences 1.6.5** and **1.6.8** have the form of propositional sentences but it is not clear that they express propositions.

Sentence 1.6.5 appears declarative, yet it refers to a non-existent entity, the current king of France, so it is not clear whether we should regard it as being false or meaningless. In the latter case, it would not express a proposition.

Similarly, **Sentence 1.6.8** makes a claim about itself. It is not clear that sentences as such should be regarded as meaningful at all. On the other hand, it seems clear that the sentence is saying something true. Thus, whether this counts as a propositional sentence or not will depend on philosophical considerations.

Solution to task 1.6.

1. Propositional sentence (proposition about the Solar system)
2. Interrogative sentence
3. Propositional sentence (proposition about geometry)
4. Imperative sentence
5. Sentence of declarative form, which may express a proposition depending on our interpretation of it.
6. Propositional sentence (conditional proposition)
7. Propositional sentence (false proposition about the world)
8. Sentence of declarative form, which may express a proposition depending how we understand truth.

True and false sentences. The sentence-proposition distinction is philosophically important, and becomes even more subtle if we introduce the notion of a *statement*. Important as these distinctions are, however, they will not play a role in this textbook.

For convenience, we will also say that a sentence can be true or false given the conditions stated below.

Notion 1.14 (True and false sentence). A propositional sentence is true if the proposition it stands for is true. A propositional sentence is false if the proposition it stands for is false.

This approach is useful, as the concepts of truth and falsehood will be defined in relation to formulae in a formal language, which are the analogues of sentences in natural language.

Note that the statement of **notion 1.14** is compatible with there being a sentence of declarative form which is neither true nor false, as it would fail to express a proposition.

Task 1.7. Classify the following sentences into: true (t), false (f), and expressing no proposition (n).

1. Paris is the capital of France.
2. The Earth is flat.
3. Some triangles have four sides.
4. Bruce Wayne is Batman.
5. Bruce Wayne is Superman.
6. The round square is a square that is round.
7. There is a present king of France.
8. The present king of France is bald.
9. This sentence is in English.
10. Colourless green ideas sleep furiously.
11. If humans are mortal, then they are mortal.
12. If humans are mortal, then they are not.
13. Either humans are mortal or are not mortal.
14. Humans are mortal and are not mortal.
15. This sentence has exactly six words.
16. This sentence has more than ten words.
17. This sentence is true.
18. This sentence is false.
19. This sentence expresses a proposition.
20. This sentence expresses no proposition.

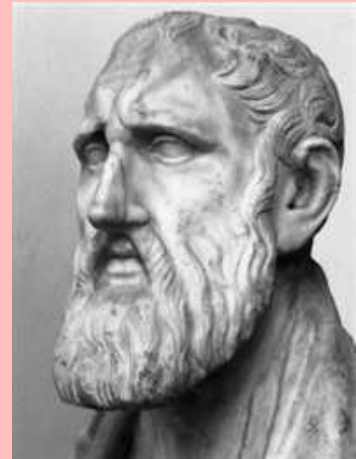
Some sentences in **task 1.7** clearly express propositions, and these propositions can be straightforwardly evaluated as true or false; for instance, **sentences 1.7.1, 1.7.2, and 1.7.3**.

Some other sentences are true or false logic. For instance, the **sentence 1.7.13** is true, since it exhausts the only two possibilities concerning humanity and mortality, while **sentence 1.7.14** is false because it asserts and denies the same property simultaneously. Likewise, **sentence 1.7.11** is logically true. As for **sentence 1.7.12**, it is debatable whether it is logically false. However, it happens to be

Historical note: The sayables

The Stoic concept of the *sayable* (λεκτόν) is probably the earliest antecedent of our modern notion of a proposition. Sextus Empiricus (ca. 160–ca. 210) reports about this in his *Πρὸς λογικούς* (*Against the Logicians*, 170–ca. 190, VII, 11–2):

some placed the true and false in the region of the thing signified, others in that of utterance, others in that of the motion of thought. And the Stoics stood for the first opinion, saying that three things were connected with one another, the thing signified and the signifier and the object. Of these the signifier is the utterance (e.g., the utterance “Dion”); the thing signified is the actual state of affairs revealed by it, and which we apprehend as it subsists in our thought ...; and the object is the externally existing thing (e.g., Dion himself). And of these, two are bodies, i.e. the utterance and the object, while one is incorporeal, i.e. the state of affairs signified and sayable, which is true or false.



Zeno of Citium
(ca. 334–ca. 262 BCE),
founder of Stoicism

false, because humans are indeed mortal, and it does not follow from this that they are also not mortal.

Next we have some sentences expressing descriptions about themselves, as expressions. For example, we can see [sentences 1.7.15](#) and [1.7.16](#) are true and false, respectively, if we count the words in the sentences themselves. [Sentence 1.7.9](#) is true for similar reasons, as it correctly describes the language in which it is written.

Now we find sentences concerning fictional and impossible objects. These sentences either fail to express a proposition, or are true or false depending on the properties such object would have in their respective fictional and impossible worlds. Thus, as hopefully most readers know, [sentences 1.7.4](#) and [1.7.5](#) would be true and false, respectively.[†] As for [sentence 1.7.6](#), it is quite difficult to think of a world where a round square can exist. But if it does, [sentence 1.7.6](#) would be true in it. We might also say that the sentence is just false, as it states the existence of an impossible entity.

We also have a couple of sentences involving – not impossible but – non-existent entities. Thus, [sentence 1.7.7](#) is false, as it clearly states that a king of France currently exists, which is not the case. Instead, [sentence 1.7.8](#) does not openly say this, but seems to presuppose it. As with [sentence 1.7.6](#), if we understand this as implying that a current king of France is said to exist, then [sentence 1.7.8](#) is false. Else, these sentences fail to express propositions.

[Sentence 1.7.10](#) is a clear case of a sentence failing to express a proposition. [Sentences 1.7.17–1.7.20](#) also seem to fall within this group. However, if you think about them, you might see that they may be regarded true or false. After all, if [sentences 1.7.16](#) and [1.7.16](#) express propositions, why cannot [sentences 1.7.17–1.7.20](#)?

[†]: However, see J. M. DeMatteis' *Superman: Speeding Bullets* (DC Comics, Elseworlds, 1993). In this fictional world, Bruce Wayne might be said to be Superman.

1.4 Placeholders and Quasi-Quotation

Solution to task 1.8.

t: 1, 9, 11, 13, 15

f: 2, 3, 7, 12, 14, 16

n: 10

t or n: 4

f or n: 5, 6, 8

n, or neither t nor f: 17, 19

n, or both t and f: 18, 20

∗: Most authors use, depending on the kind of expression, a specific selection of italicised Roman or Greek letters, such as: ‘A’, ‘v’, ‘φ’, ‘ψ’, ‘μ’. See, for example, the historical note on page 21. In this case, we usually call these expressions *meta-variables*.

†: Make sure you understand that it is the symbols within the quote marks – rather than these symbols plus the quote marks – that will serve as placeholders, and why.

Placeholders. We have introduced a simple device to speak about any *specific expression* we want. With it, we can say things like: ‘dog’ and ‘kid’ are nouns in English; ‘tiger’ has five characters and “tiger” has seven; etc. This device, however, is not enough for speaking about an *unspecified expression*.

For instance, we may want to speak about an expression *e* which has five letters and three numerals, or about a word *w* with two consonants and three syllables, without specifying an exact expression or word having those features. Rather, we want to speak generally about expressions having those features. In both cases, ‘*e*’ and ‘*w*’ serve as *expression placeholders* – or *placeholders*, for short.

Notion 1.15 (Placeholder). An **expression placeholder** is an expression which stands for an unspecified expression.

So, if we use a placeholder, like ‘*w*’ above, to speak about an unspecified word, we say that ‘*w*’ stands for a word, or that *w* is a word – in the latter case, we do not use quote marks.

We will use a specific selection of symbols as placeholders.∗ We introduce, first, a special kind of placeholders.

Notation 1.16 (Basic placeholder). The type-writer font letters:

‘A’, ‘B’, ‘C’, ‘D’, ‘E’, ‘F’, ‘G’, ‘H’, ‘I’, ‘J’, ‘K’, ‘L’, ‘M’,
‘N’, ‘O’, ‘P’, ‘Q’, ‘R’, ‘S’, ‘T’, ‘U’, ‘V’, ‘W’, ‘X’, ‘Y’, ‘Z’,
‘a’, ‘b’, ‘c’, ‘d’, ‘e’, ‘f’, ‘g’, ‘h’, ‘i’, ‘j’, ‘k’, ‘l’, ‘m’,
‘n’, ‘o’, ‘p’, ‘q’, ‘r’, ‘s’, ‘t’, ‘u’, ‘v’, ‘w’, ‘x’, ‘y’, ‘z’,

as well as:

‘1’, ‘2’, ‘3’, ‘4’, ‘5’, ‘6’, ‘7’, ‘8’, ‘9’, ‘0’,
‘1’, ‘2’, ‘3’, ‘4’, ‘5’, ‘6’, ‘7’, ‘8’, ‘9’, ‘0’,

will serve as **basic placeholders** and, hence, as **placeholders**.[†]

Having already some placeholders available, we may reformulate the statement of **notation 1.8** more briefly and nicely as follows:

Fact 1.17. *e* can be mentioned via an expression formed by: first, a left quote mark, next, *e*, and, finally, a right quote mark.

However, it would be a mistake to reformulate it as follows:

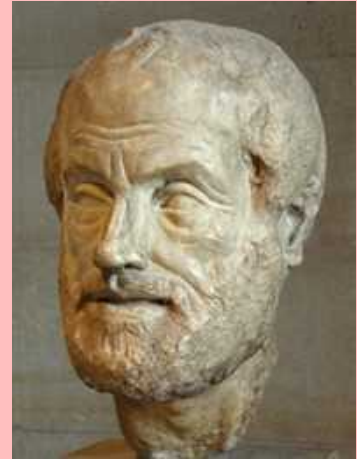
Mistake 1.18. An expression *e* can be mentioned via ‘*e*’.

In **mistake 1.18**, the first occurrence of ‘*e*’ functions as a placeholder for an unspecified expression. Its second occurrence, however,

Historical note: Schematic letters

Placeholders originate in schematic letters, which function in much the same way as our placeholders. The earliest recorded use of such devices is found in the logical works of Aristotle. See, for instance, the following passage of his *Ἀναλυτικὰ Πρώτερα* (*Prior Analytics*, ca. 351 BCE, I 2, 25a20–26):

if A belongs to some of the Bs, then necessarily B belongs to some of the As. (For if it belongs to none, then neither will A belong to any of the Bs.) But if A does not belong to some B, it is not necessary for B also not to belong to some A (e.g. if B is animal and A man: for man does not belong to every animal, but animal belongs to every man).



Aristotle (384–322 BCE)

appears within quote marks in “e”, which mentions the very variable ‘e’ rather than an unspecified quoted expression.

Task 1.8. Let *i* be an unspecified adjective containing at least one vowel and at most three consonants. What of the following is true, false, or undetermined about *a* and ‘*a*’?

- | | |
|---|--|
| 1. <i>a</i> is a word. | 15. <i>a</i> stands for a long word. |
| 2. ‘ <i>a</i> ’ is a word. | 16. ‘ <i>a</i> ’ stands for a short word. |
| 3. <i>a</i> is a long word. | 17. <i>a</i> is a numeral. |
| 4. ‘ <i>a</i> ’ is a long word. | 18. ‘ <i>a</i> ’ is a numeral. |
| 5. <i>a</i> is a short word. | 19. <i>a</i> stands for a numeral. |
| 6. ‘ <i>a</i> ’ is a short word. | 20. ‘ <i>a</i> ’ stands for a numeral. |
| 7. <i>a</i> has at least one vowel. | 21. <i>a</i> is a placeholder. |
| 8. ‘ <i>a</i> ’ has at least one vowel. | 22. ‘ <i>a</i> ’ is a placeholder. |
| 9. <i>a</i> has at least two vowels. | 23. <i>a</i> stands for a placeholder. |
| 10. ‘ <i>a</i> ’ has at least two vowels. | 24. ‘ <i>a</i> ’ stands for a placeholder. |
| 11. <i>a</i> has four consonants. | 25. “ <i>a</i> ” stands for a placeholder. |
| 12. ‘ <i>a</i> ’ has four consonants. | 26. “ <i>a</i> ” stands for an expression that stands for a placeholder. |
| 13. <i>a</i> stands for a word. | |
| 14. ‘ <i>a</i> ’ stands for a word. | |

First, since *a* is stipulated to be an adjective, which means it is a word. Hence, 1.8.1 is true. By contrast, ‘*a*’ is not a word but a placeholder, so 1.8.2 is false.

Next, we cannot know how large *a* is from the given information that it contains at least one vowel and at most three consonants. Hence, it is undetermined whether 1.8.3 and 1.8.5 are true. With regard to 'a', as it is not a word, 1.8.4 and 1.8.6 are just false.

By assumption, *a* has at least one vowel, so 1.8.7 is true. As for 'a', it in fact contains exactly one vowel, namely 'a', so 1.8.8 is also true. As for 1.8.10, 'a' clearly does not have two vowels, making it false. The placeholder 'a', however, contains no consonants at all, so 1.8.12 is false. Now, 1.8.9 is undetermined, as we do not know that *a* has more than one vowels – although it is likely, given that it has three consonants. However, 1.8.11 is false, since *a* has at most three consonants, which makes it impossible for it to have four.

Now, being *a* an adjective, it does not stand for a word – it rather stands for something like a *property*. Therefore, 1.8.13 and 1.8.15 are false. By contrast, 'a' is a placeholder for an adjective, so it stands for an adjective and, hence, for a word, which makes 1.8.14 true. Whether 'a' stands for a short word cannot be determined with the information given, which makes 1.8.16 undetermined.

Since *a* is an adjective, it is neither a numeral nor a placeholder, so 1.8.17 and 1.8.21 are false; nor does it stand for a numeral or a placeholder, so 1.8.19 and 1.8.23 are also false. For the same reason, 'a' stands for an adjective, which means it does not stand for a placeholder or for a numeral, so 1.8.24 and 1.8.20 are false. Now, since 'a' is a placeholder, it is not a numeral, so 1.8.22 is true and 1.8.18 is false.

Finally, we mention expressions in quote marks, like "a", which is obtained by placing quote marks around 'a'. Thus, "a" mentions and, hence, stands for the placeholder 'a', which makes 1.8.25. However, "a" stands for 'a', which is a placeholder, but which does not stand for a placeholder. Hence, 1.8.26 is false.

Solution to task 1.8.

True: 1, 7, 8, 14, 22, 25.

False: 2, 4, 5, 10, 11–15, 17–21, 23, 24, 26.

Undetermined: 3, 6, 9, 16.

Quasi-quotation. Suppose we wish to speak about an unspecified letter enclosed in braces, like '{A}', '{a}', '{B}', '{b}', etc. Previous to introducing our quote marks notation, we might describe it as an expression of the form:

$\{L\}$, where *L* is a letter.

However, it is not completely clear what are the braces doing there. They may be part of the form of the expression, as we wish. However, we could also read them as delimiting the very form of the expression, so that it is clear that the comma is not part of it.

Using quote marks, we might try again and describe our expression as having the form:

'{L}', where *L* is a letter.

Historical note: Quasi-Quotation

Quasi-quotation was introduced by Willard Van Orman Quine in his book *Mathematical Logic* (1940, pp. 35–6) as follows:

There is need also of a convenient way of speaking of specific contexts of unspecified expressions: speaking, e.g., of the result of enclosing the unspecified expression μ in parentheses... Note that quotation is not available here. The quotation... ' (μ) ' designate[s] only the specific expression[] therein depicted, containing the specific Greek letter[] ' μ '... Reference to the intended contexts of the unspecified expressions ... will be accomplished by a new notation of *corners*, thus $\ulcorner(\mu)\urcorner$. Because of the close relationship which it bears to quotation, this device may be called *quasi-quotation*. It amounts to quoting the constant contextual background[], ' $(\)$ ' ... and imagining the unspecified expression[] μ ... written in the blanks.



W. V. O. Quine (1908–2000)

But this is even worst, since " $\{L\}$ " mentions a specific expression – which contains the specific symbol 'L'. It does not denote an unspecified letter occurring between braces. In order to denote this, we need a more flexible kind of quotation.

Notation 1.19 (Quasi-quotation). An expression formed by:

first, ' $\ulcorner$ ', next, e, and, finally, ' $\urcorner$ '

functions as the expression resulting from enclosing e in simple quote marks, but with any placeholder occurring in e still functioning as a placeholder.

In other words, *quasi-quotation* behaves like normal quotation, except that occurrences of placeholder expressions inside it are not taken as mentions of the placeholder itself. Instead, they maintain their condition of placeholders, thus standing for unspecified expressions within the quasi-quoted expression.

In particular, when a placeholder occurs alone inside a quasi-quotation, as in ' $\ulcorner e \urcorner$ ', the effect is exactly the same as using the placeholder 'e' itself. In such a case, the quasi-quotation adds nothing: it simply stands for whatever expression 'e' stands for.

Using this notation, we may refer to our unspecified letter enclosed in braces as an expression of the form:

$\ulcorner\{L\}\urcorner$, where L is a letter. ($\urcorner$)

Thus, in ' $\ulcorner\{L\}\urcorner$ ', the symbol 'L' is not being mentioned – if anything, we are mentioning an arbitrary letter L. Therefore, our quasi-quotation ' $\ulcorner\{L\}\urcorner$ ' succeeds to mention an unspecified expression enclosed in braces $\ulcorner\{L\}\urcorner$. The further condition that L be a letter makes ($\urcorner$) mention an arbitrary letter enclosed in braces.

Task 1.9. Rewrite the expression form described by each item below using placeholders and quasi-quotation.

1. Two letters connected by a hyphen, i.e., by ‘-’.
2. One numeral and one letter, in that order.
3. A lower-case letter, two upper-case letters, and an asterisk, in that order.
4. A lower-case letter, two upper-case letters, and any expression, in that order.
5. Expressions representing basic arithmetic operations (addition, subtraction, multiplication, and division) between natural numbers.
6. Expressions representing arbitrary mathematical operations between the numbers 1 and 2, in that order.
7. Expressions representing basic addition and multiplication between the numbers 1 or 2.
8. Categorical statements, that is, expressions of the form: category-term copula category-term, such as ‘Man is mortal’.

For each placeholder you use, specify whether it stands for a letter, a numeral, a digit, a category term, an operator, etc.

In the first four tasks, we have simple sequences of letters, numerals, or symbols. In [task 1.9.1](#), we have two letters l and m connected by ‘-’, so its form is $\ulcorner l-m \urcorner$. In [task 1.9.2](#), a numeral n precedes a letter l , so its form is $\ulcorner nl \urcorner$. In [task 1.9.3](#), we have a lower-case letter a followed by two upper-case letters A , B , and ‘*’, so its form is $\ulcorner aAB* \urcorner$. [Task 1.9.4](#) is similar, but we have an arbitrary expression $\bullet$ at the end instead of ‘*’, so its form is $\ulcorner aAB\bullet \urcorner$.

Note that, in [tasks 1.9.2](#) and [1.9.4](#), the corner marks are optional, since they enclose only placeholders.

In the next three of tasks, we have mathematical expressions of the form $\ulcorner a \heartsuit b \urcorner$, with different conditions on their placeholders. In [task 1.9.5](#), a and b are numerals standing for natural numbers, and $\heartsuit$ is either ‘+’, ‘-’, ‘ $\times$ ’, or ‘ $\div$ ’.[‡] In [task 1.9.6](#), $\heartsuit$ is a mathematical operator, a is the numeral ‘1’ and b is ‘2’. Hence, our expression is also of the form $\ulcorner 1 \heartsuit 2 \urcorner$, where $\heartsuit$ is a mathematical operator. In [task 1.9.7](#), a and b are either ‘1’ or ‘2’, and $\heartsuit$ is either ‘+’ or ‘ $\times$ ’.

Finally, [task 1.9.8](#) presents the type of expression that has been most widely studied throughout history: categorical expressions, whose form is $\ulcorner s \heartsuit p \urcorner$, where s and p are terms, and $\heartsuit$ is a copula.

Note that, in [tasks 1.9.5–8](#), the corner marks are not optional, as there are spaces contained within them. In [1.9.8](#) we have two occurrences of the normal space ‘ ’, while in the others we have two occurrences of the thin space ‘ ’.

[‡]: Hence, we might also say that our expression is of one of the forms $\ulcorner a + b \urcorner$, $\ulcorner a - b \urcorner$, $\ulcorner a \times b \urcorner$, $\ulcorner a \div b \urcorner$.

Solution to task 1.9.

The form of the expression in each subtask above is:

1. $\lceil l-m \rceil$, where l and m are letters.
2. $n1$ (or, what is the same, $\lceil n1 \rceil$), where n is a numeral and 1 is a letter.
3. $\lceil aAB^* \rceil$, where a is a lower-case letter, and A, B, C are upper-case letters.
4. $aAB\emptyset$ (or $\lceil aAB\emptyset \rceil$), where a is a lower-case letter, and A, B are upper-case letters.
5. $\lceil a \heartsuit b \rceil$, where a, b are numerals standing for natural numbers and $\heartsuit$ is '+', '-', '×', or '÷'.
6. $\lceil a \heartsuit b \rceil$, where $\heartsuit$ is a mathematical operator, a is (the numeral) '1', and b is '2'; or
 $\lceil 1 \heartsuit 2 \rceil$, where $\heartsuit$ is a mathematical operator.
7. $\lceil a \heartsuit b \rceil$, where a, b are either '1' or '2', and $\heartsuit$ is either '+' or '×'.
8. $\lceil s \heartsuit p \rceil$, where s, p are terms and $\heartsuit$ is a copula (note the separating spaces).

Note the spaces contained within 5–8.

Limit cases of placeholders. You may have noticed that, in [task 1.9.6](#), we describe a as *the* numeral '1' and b as *the* numeral '2', which are quite specific expressions. This seems to contradict the idea that placeholders stand for unspecified expressions.

This, however, is not problematic: it represents a limit case in the use of a placeholder. There are two such limit cases to consider. On one extreme, a placeholder may have no restriction, standing for any expression whatsoever. On the other extreme, a placeholder may be so restricted that it can take exactly one value, as in [task 1.9.6](#). In [task 1.9.7](#), e.g., we approach this latter limit, as a and b are restricted to only two possibilities, namely '1' and '2'.

Note, however, it is not the placeholder itself that is making the specification. Rather, it is the further conditions we provide about

Historical note: Restrictions

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Euclid (XXXX–XXXX)

it that imposes it. Hence, even in this limit cases, placeholders remain expressions standing for unspecified expressions.

1.5 Schema and Instance

Schema and instance. An expression with placeholders is no *specific expression*. It is rather kind of a template or *schema* from which specific expressions may be obtained.

Notion 1.20 (Specific expression). A **specific expression** is an expression in which no placeholder occurs.

Notion 1.21 (Schematic expression). A **schematic expression or schema** is an expression in which at least one placeholder occurs, which may be restricted to a certain type of expressions.

Consider, for example, the schema:

‘the father of n’ (T)

From this, we can obtain specific expressions such as:

‘the father of Plato’, (O)

‘the father of Aristotle’, (II)

‘the father of the Moon’, (E)

‘the father of fo rehtaf eht’. (Ω)

*: Moreover, ‘{A}’, ‘{a}’, etc. are *instances* of schema (T) on p. 21.

Each of these expressions obtained is an *instance* of schema (T).*

Notion 1.22 (Instance). A specific expression resulting from replacing all placeholders in a schema with suitable specific expressions, where every occurrence of the same placeholder is replaced with the same expression, is an **instance** of that schema.

Our statement of **notion 1.22** prevents a schema from being an instance of another. When we consistently replace the placeholders of a schema with suitable specific expressions or schemas, the resulting schema is called a *subschema* of the original one.

Notion 1.23 (Subschema). A schema is a **subschema** of another schema when all instances of the former are also instances of the latter. Moreover, it is a **proper subschema** when the latter has at least one instance which is not an instance of the former.

†: Plural of ‘schema’.

It trivially follows that all schemata[†] are subschemata of themselves, but not proper subschemata of themselves.

Historical note: Schemata

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX–XXXX)

Fact 1.24. Each schema is a subschema of itself.

Fact 1.25. No schema is a proper subschema of itself.

Now, in specifying a schema such as (Υ) , we would normally want (♂) and (♀) to count as instances, but not (☾) and (♂♂) . We can do so by further specifying the schema, so that restrictions are placed on its placeholders.

Consider, for instance, the following variation of schema (Υ) :

‘the father of n ’, where n is the name of a person. (Υ')

Note that (♀) and (♂) are instances of schema (Υ) as much as of (Υ') . Expressions (☾) and (♂♂) , instead, do not count as instances of (Υ') , since ‘the Moon’ and ‘fo rehtaf eht’ are not names of persons. Also note that schema (Υ') is a subschema of (Υ) .

To make schemata easier to state, we fix the following notation.

Notation 1.26. ‘ $E : c$ ’ expresses a schema whose instances are all those expressions resulting from replacing the placeholders in ‘ E ’ with expressions satisfying the conditions stated by ‘ c ’.

Thus, we might rewrite schema (Υ') equivalently as follows:

the father of $n : n$ is the name of a person (Υ'')

Note that we do not enclose the expression to the left of ‘:’ in corner marks as such quasi-quotation is here implicit.

Schemata are very useful devices for speaking about the form common to two or more expressions, which may be different.

Consider, for instance, the sentences:

Peter is clumsy.
Nancy is smart. (⌈⌋)

These two sentences are instances of the sentence schema:

n is a . : n is a proper name, a is an adjective (⌊⌋)

Note, however, that given the conditions stated to the right of ':', the following do not qualify as instances of (⌊⌋):

Plato is Aristocles.
Martin is.
Nancy is not clumsy.
Smart is Nancy. (⌈⌋)

Note that, although expressions (⌈⌋) and (⌊⌋) have the same meaning, only the former is an instance of schema (⌊⌋). This is because an expression is an instance of a schema not by virtue of its meaning, but solely by virtue of its linguistic form.

Task 1.10 (Schemata). Consider the following schemata:

- e loves f . : e and f are any expression (a)
- e loves f . : e and f are proper names (b)
- e loves f . : e is a human name, f is a deity name (c)
- e loves e . : e is any expression (d)
- e loves e . : e is a proper name (e)
- e loves e . : e is a human name or a deity name (f)

For each schema above, determine:

- (i) whether it is a subschema of (a)–(f); and
- (ii) whether any of the expressions below is an instance of it.[‡]

[‡]: All human names and deity names are proper names. Moreover, although some deity names, like 'Zeus', may sometimes be used as human names, we will treat them as if they were not human names.

- | | |
|--------------------------|-----------------------|
| 1. Peter loves Laura | 6. She loves you |
| 2. Peter loves Peter | 7. Odin loves music |
| 3. Horus loves Horus | 8. music loves music |
| 4. Today loves yesterday | 9. And I love her |
| 5. Horus loves Anna | 10. She loves herself |

First, note that all schemata are subschemata of (a). Hence, all expressions but 1.10.9 are instances of schema (a), because instead of the inflection 'loves', which is required by our schema, we have 'love'. In fact, 1.10.9 is an instance of no schema above.

Second, no expression with different expressions to the left and right of 'loves' is an instance of schemata (d)–(f). This includes 1.10.10, even though 'she' and 'herself' share the same referent.

Third, (d) and (e) are proper subschemata of (a) and (b), respectively. Hence, all instances of schemata (d) and (e) are also instances of (a) and (b), respectively, but not the other way around.

Finally, however, schema (f) is not a subschema of (c), as the latter's placeholders have different conditions. Hence, not all instances of (f) are instances of (c). This is the reason why 1.10.5 is not an instance of schema (c).

Solution to task 1.10.

(i) Each cell indicates whether the schema listed in the first column of its row is a subschema of the schema listed in the top row of its column.

	(a)	(b)	(c)	(d)	(e)	(f)
(a)	Yes	No	No	No	No	No
(b)	Yes	Yes	No	No	No	No
(c)	Yes	Yes	Yes	No	No	No
(d)	Yes	No	No	Yes	No	No
(e)	Yes	Yes	No	Yes	Yes	No
(f)	Yes	Yes	No	Yes	Yes	Yes

A schema indicated above as subschema of another schema, is also its proper subschema.

(ii) Each cell indicates whether a numbered expression is an instance of a schema.

	(a)	(b)	(c)	(d)	(e)	(f)
1	Yes	Yes	No	No	No	No
2	Yes	Yes	No	Yes	Yes	Yes
3	Yes	Yes	No	Yes	Yes	Yes
4	Yes	No	No	No	No	No
5	Yes	Yes	No	No	No	No
6	Yes	No	No	No	No	No
7	Yes	No	No	No	No	No
8	Yes	No	No	Yes	No	No
9	No	No	No	No	No	No
10	Yes	No	No	No	No	No

Expressing schemata. There is an important difference between specific expressions and schemata in terms of their meaning.

A specific expression is either meaningful or not. If it is meaningful, it can stand for a specific referent – although it might be philosophically problematic to determine it. Thus, a specific expression either is meaningful and stands for its referent or it is unmeaningful and stands for nothing.

To exemplify, consider again the following expressions:

'the father of Plato',	(ϝ)
'the father of Aristotle',	(ϝ)
'the father of the Moon',	(Ϟ)
'the father of fo rehtaf eht'.	(Ϟ)

Expressions (ϝ) and (ϝ) stand for something, Aristotle's father (i.e., Nicomachus) and Plato's father (i.e., Ariston), respectively. Expressions (Ϟ) and (Ϟ), instead, lack meaning – at least a literal one – in English and, hence, have no referent.

A schema, on the other hand, though it may be regarded as meaningful in some sense, cannot stand for a specific referent in the way a specific expression can. Now, as we saw above, we can

Historical note: Schematic axioms

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX–XXXX)

make a schema more specific the more restrictions we place on its placeholders, as we did in (Υ'').

the father of n : where n is the name of a person (Υ''')

Of course, even with these restrictions, schema (Υ''') does not succeed to stand for a specific referent. Consider, however, the following schema:

the father of n is a relative of n
 : n is the name of a person. ($\times'$)

Not only is it clear that schema ($\times'$) is in some sense meaningful, but it also seems to have some kind of referent. Whoever utters ($\times'$) seems to be stating that the father of a person (with a name) is a relative of that very person, which is indeed true.

Schema ($\times'$) expresses each expression obtained by replacing ' n ' with a proper name in 'the father of n is a relative of n '. This might idea be generalised as follows.

Notion 1.27 (Expressing a schema). To **express a schema** functions as expressing each of its instances.

In other words, stating a schema is a way of expressing many – often infinitely many – things at once.

However, this does not mean, e.g., that a sentence schema is true when all its instances are true and false otherwise. A sentence schema cannot be said to be true or false. All we can say is that all or some of its instances are true or false.

Open expressions. Consider the following schema:

$$n \times n = 2 \times n : n \text{ stands for a number.} \quad (\textcircled{\text{f}})$$

Since 'n' stands for a number, schema $(\textcircled{\text{f}})$ functions as expressing, of each number, that multiplying it by itself equals twice that number – which is not true in general. However, a mathematician would not express this in this way, but rather as follows:

$$x \times x = 2x, \text{ where } x \text{ is a number.} \quad (\textcircled{\text{w}})$$

Note that, in schema $(\textcircled{\text{f}})$, 'x' is a number variable, i.e., it stands for a number. In $(\textcircled{\text{w}})$, instead, 'n' is a placeholder standing for a number expression. Hence, while $(\textcircled{\text{f}})$ is schematic in the sense of **notion 1.21**, formula $(\textcircled{\text{w}})$ is not, it is a specific expression.

However, $(\textcircled{\text{w}})$ may be said to be schematic in another sense. First, note that 'x' is an expression that stands for an unspecified number, that is, is a number variable.

Notion 1.28 (Variable). A **variable** is a symbol that serves as a placeholder for arbitrary elements of a given domain.

Now, the rules of the mathematical system in which $(\textcircled{\text{f}})$ can be expressed should allow us to replace 'x' with any expression that stands for a number. Specific expressions containing variables are not quite common in natural language, but are more proper of mathematics.

Sometimes, we make the variables specific enough for the expression to have a specific meaning, as when we say:

$$\text{For all numbers } x, x \times x = 2x. \quad (\textcircled{\text{v}})$$

$$\text{For some number } x, x \times x = 2x. \quad (\textcircled{\text{v}})$$

Thus, $(\textcircled{\text{v}})$ is false, since 3 is a number and it is not the case that $3 \times 3 = 2 \times 3$. On the other hand, $(\textcircled{\text{v}})$, since 0 is a number, and it is the case that $0 \times 0 = 2 \times 0$.

What about $(\textcircled{\text{w}})$? Can we say of this expression that it is true or false? Not quite, since, although $(\textcircled{\text{w}})$ is a specific expression, it does not stand for a specific proposition, but for an arbitrary one of a certain form. Expressions of these kinds are *open expressions*.

Notion 1.29 (Open expression). An **open expression** is a specific expression which does not have a specific referent, but an arbitrary one within some constraints.

Thus, $(\textcircled{\text{w}})$ is an open expression whose referent is an arbitrary proposition of the form ' $n \times n = 2n$ ', where n stands for a number.

Historical note: Open expressions

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX-XXXX)

This shows the sense in which ($\approx$), and open expressions in general, may also be thought of as schemata. For this reason, we will also use **notation 1.26** for representing open expressions.

Notation 1.30. $\ulcorner E : c \urcorner$ expresses an open expression whose variables are constrained by the conditions stated by $\ulcorner c \urcorner$.

Thus, ($\approx$) may be equivalently expressed as follows:

$$x \times x = 2x : x \text{ is a number.}$$

1.6 Definiendum and Definiens

Definitions. So far, we have been introducing some notions and notation but in a not very rigorous way. In fact, we lack a notation for introducing notions and notations rigorously. In order to address this, we will introduce a concept and a notation for *definitions*.

Notion 1.31 (Definition). A **definition** is an expression that stipulates the meaning of a yet unmeaningful term as used within an expression, the **definiendum**, to be equal to the meaning of an already meaningful expression, the **definiens**.

Note that the term being defined is not necessarily the same as the definiendum, as we will see later.

In this textbook, we will take for *an already meaningful expression* one whose meaning we can understand by virtue a notion, a notation, or a definition previously stated. Definitions will always be introduced in their standard notation.

Notation 1.32 (Standard form of definitions). $\ulcorner U \stackrel{\text{DF}}{=} M \urcorner$ expresses a definition whose definiendum is U and whose definiens is M.*

*: As with [notation 1.26](#), we need not enclose the definiendum or the definiens in corner or quote marks, as these are implicit.

For instance, Aristotle is not only known by the name ‘Aristotle’, but also with other names, like for instance, ‘the Stagirite’. Assuming that we know the meaning of the former but not of the latter, we might establish the meaning of ‘the Stagirite’ via:

$$\text{the Stagirite} \stackrel{\text{DF}}{=} \text{Aristotle} \quad (\Uparrow)$$

Definition $(\Uparrow)$ does not quite say that Aristotle is the same person as the Stagirite. It rather stipulates the meaning of ‘the Stagirite’ to be the same as the meaning of ‘Aristotle’. In other words, definition $(\Uparrow)$ stipulates that, by using ‘the Stagirite’ we mention the same person we would mention by using ‘Aristotle’, i.e., Aristotle.

Fact 1.33. The definiens of a definition is not stipulated to be the meaning of the definiendum, but rather to express the same meaning as the definiendum.

Once a definition is stipulated, the definiendum and definiens become interchangeable under some conditions.

Fact 1.34 (Definiendum introduction). Let $\ulcorner U \stackrel{\text{DF}}{=} M \urcorner$ be a stated definition and e an expression where M is used. If i is an expression obtained from e by replacing one or more occurrences of M among those occurrences in which M is used (rather than mentioned) with corresponding occurrences of U, then e and i have the same meaning.

Historical note: Definiendum and definiens

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX–XXXX)

Fact 1.35 (Definiendum elimination). Let $\ulcorner U \stackrel{\text{DF}}{=} M \urcorner$ be a stated definition and e an expression where U is used. If i is an expression obtained from e by replacing one or more occurrences of U among those occurrences in which U is used (rather than mentioned) with corresponding occurrences of M , then e and i have the same meaning.

The condition that the occurrence of M be used rather than mentioned prevents us from replacing expressions within quote marks. This is essential, since those replacements could result in expressions with different meanings.

Thus, for instance, the following two sentences have the exact same meaning by virtue of definition ($\ulcorner$).

Plato was the master of the Stagirite. ($\textcircled{\text{S}}$)

Plato was the master of Aristotle. ($\textcircled{\text{A}}$)

However, note that the following two expressions do not have the same meaning even after the stipulation of definition ($\ulcorner$).

The word ‘the Stagirite’ has nine letters. ($\textcircled{\text{S}}$)

The word ‘Aristotle’ has nine letters. ($\textcircled{\text{A}}$)

The reason is that, whereas the expressions in the definiendum and definiens of ($\ulcorner$) are used in ($\textcircled{\text{S}}$) and ($\textcircled{\text{A}}$), respectively, they are not in used in neither ($\textcircled{\text{S}}$) nor ($\textcircled{\text{A}}$), but just mentioned. A fact which corroborates that latter two sentences do not have the same meaning is that ($\textcircled{\text{S}}$) is false, while ($\textcircled{\text{A}}$) is true.

To finish, note that an already stated definition has already fixed the meaning of its definiendum. Hence, it can no longer be redefined.

Fact 1.36. Once a definition has been stated, its definiendum cannot be used as the definiendum of a new definition.

Schematic definitions. Definition ($\Uparrow$) is fairly simple, as it just stipulates the equivalence in meaning of two quite specific expressions. We may nevertheless want to define not just a specific expression, but of many expressions of the same form.

Suppose, for instance, that we want to define ‘complex’ as ‘not simple’. Following the pattern of ($\Uparrow$), we may try:

$$\text{complex} \stackrel{\text{DF}}{=} \text{not simple} \quad (\mathbb{M})$$

Thus, the following expressions would have the same meaning:

Logic is complex.
Logic is not simple.

But so would the following expressions, which we did not intend:

2i is complex.
2i is not simple.

Clearly, these two sentences differ in meaning because ‘complex’ has a different sense when combined with numerals than when combined with names of subjects. This latter sense is the one we wanted to capture with ($\mathbb{M}$), and it can only be established if we somehow restrict the expressions it may be combined with.

In order to do so, we need to employ a special kind of definition.

Notion 1.37 (Schematic definition). A **schematic definition** is a schema all of whose instances are definitions.

Following **notion 1.27**, to state a definition schema functions as stating all the definitions which are its instances.

Thus, with a schematic definition, we can stipulate expressions of the form ‘s is complex’ to mean the same as ‘s is not simple’, where s stands for a subject matter. We do this as follows:

$$\begin{aligned} \text{s is complex} &\stackrel{\text{DF}}{=} \text{s is not simple} \\ &: \text{s stands for a subject matter} \quad (\underline{\underline{=}}) \end{aligned}$$

The following are instances of definition schema ($\underline{\underline{=}}$):

informatics is complex $\stackrel{\text{DF}}{=}$ informatics is not simple
linguistics is complex $\stackrel{\text{DF}}{=}$ linguistics is not simple
mathematics is complex $\stackrel{\text{DF}}{=}$ mathematics is not simple

In fact, definition schema ($\underline{\underline{=}}$) results in stipulating a definition of the form ‘s is complex $\stackrel{\text{DF}}{=}$ s not simple’ for each expression s

Historical note: Identity symbol

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX–XXXX)

denoting a subject. This amounts to defining ‘complex’ as ‘not simple’ just when applied to expressions naming subjects.

We can use schematic definitions where more than one placeholder occurs. In particular, it will be useful to introduce a few shorthands for common logical expressions in English.

Notation 1.38 (Abbreviation of logical expressions).

$\neg A \stackrel{\text{DF}}{=} \text{It is not the case that } A.$

$A \wedge B \stackrel{\text{DF}}{=} A \text{ and } B.$

$A \vee B \stackrel{\text{DF}}{=} A \text{ or } B \text{ or both } A \wedge B.$

$A \Rightarrow B \stackrel{\text{DF}}{=} \text{If } A, \text{ then } B.$

$A \Leftrightarrow B \stackrel{\text{DF}}{=} A \text{ if and only if } B.$

$\forall v A \stackrel{\text{DF}}{=} \text{For all } v, \text{ it is the case that } A.$

$\exists v A \stackrel{\text{DF}}{=} \text{For some } v, \text{ it is the case that } A.$

$\exists! v A \stackrel{\text{DF}}{=} \text{For exactly one } v, \text{ it is the case that } A.$

: A, B are sentences or open sentences,
 v is a variable.

†: That is, assuming the expressions and open expressions they contain are meaningful.

Thus, the following expressions are now meaningful[†]:

$\neg \text{Hank is a criminal.}$

$\text{Gus is a father} \vee \text{Mike is a grandfather.}$

$\text{Walter is a teacher} \wedge \text{Jesse is a student.}$

$\text{Lalo is the boss} \Rightarrow \neg \text{Nacho is the boss.}$

$\text{Walter is a criminal} \Leftrightarrow \text{Jesse is a criminal.}$

$\exists x \text{ Hank captured } x.$

$\forall x x \text{ knows Gus.}$

It is important to note here that the symbols ' $\Rightarrow$ ', ' $\Leftarrow$ ', ' $\mathbb{W}$ ', ' $\Rightarrow$ ', ' $\Leftrightarrow$ ', ' $\mathbb{W}$ ', and ' $\exists$ ' introduced in **notation 1.38** are not the definienda[‡] of the definitions there stipulated, but the terms defined. ‡: Plural of 'definiendum'.

Their meanings are, of course, stipulated by that definition, but by means of defining expressions of a certain form in which they occur. Therefore, **notation 1.38** does not fix the meaning of any the following expressions:

The teacher is either Ana $\mathbb{W}$ Bruce.
Jimmy $\Leftarrow$ Chuck are brothers.

Another useful abbreviation will be the following:

Notation 1.39 (Abbreviation of identity).

$A = B \stackrel{\text{DF}}{=} A$ is identical to B
 $A \neq B \stackrel{\text{DF}}{=} \neg A = B$
: A, B are meaningful expressions.

This means that '=' will be understood in the sense of expressing the referents of two given meaningful expressions are identical, or equal, or the same.

Thus, the following expressions are now meaningful:

Walter = Heisenberg.
the author of *Demian* = the author of *Der Steppenwolf*.
 $1 + 2 = 4$.

All the expressions above are specific sentences, the latter of which is false.

Other expressions equivalent to ' $A = B$ ', in some contexts, include: ' A is equal to B ', ' A equals B ', ' A is the same as B ', ' A and B are the same', ' A is B ' (an exception is when A is a subject term and B is a predicate term), among others.

Furthermore, if two expressions are stated to be equal by definition, then their referents are identical.

Fact 1.40. $A \stackrel{\text{DF}}{=} B$ is a stated definition $\Rightarrow A = B$.

Historical note: Piecewise definitions

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX–XXXX)

§: Plural of 'definiens'.

Piecewise definitions. Schematic definitions allow for definitions which establish different definientia[§] for a given term, depending on the expressions that accompany it in the definiendum. We call these *piecewise definitions*.

Notion 1.41 (Piecewise definition). A **piecewise definition** defines a term by stating different conditions under which different definientia apply.

For instance, we may want to define the expression 'n is a parent' as 'n is a father' if n stands for a man, and as meaning 'n is a mother' if n stands for a woman. We can do this by stating two separate definitions:

$$n \text{ is a parent} \stackrel{\text{DF}}{=} n \text{ is a father} : n \text{ stands for a man} \quad (\mathfrak{M}_1)$$

$$n \text{ is a parent} \stackrel{\text{DF}}{=} n \text{ is a mother} : n \text{ stands for a woman} \quad (\mathfrak{M}_2)$$

The parallel statement of definitions $(\mathfrak{M}_1)$ and $(\mathfrak{M}_2)$ does not contradict **fact 1.36** in any way. Note that $(\mathfrak{M}_1)$ and $(\mathfrak{M}_2)$ have no instance in common, so they do not assign different definientia to the same definiens.

Piecewise functions are also useful for defining some well-known mathematical functions. For instance, the following defines the absolute value function:

$$|n| \stackrel{\text{DF}}{=} n : n \text{ stands for a number greater or equal than } 0, \quad (\times_1)$$

$$|n| \stackrel{\text{DF}}{=} -n : n \text{ stands for a number smaller than } 0. \quad (\times_2)$$

Again, note that the definitions $(\times_1)$ and $(\times_2)$ have no instances in common, so there is no conflict with **fact 1.36**.

To close, note that our piecewise definition $(\times)$ is not expressed as mathematical definitions usually are. In a normal mathematical

definition, we would use open expressions like the following:

$$\begin{aligned} |x| &\stackrel{\text{DF}}{=} x && : x \geq 0, && (\mathcal{D}_1) \\ |x| &\stackrel{\text{DF}}{=} -x && : x < 0. && (\mathcal{D}_2) \end{aligned}$$

Since 'x' is a number variable, the rules of the mathematical system for which this expression is defined should allow us to replace 'x' with any expression standing for a number that satisfies the conditions stated after ':'.

Explicit and recursive definitions. In all definitions presented so far, a new term is used within the definiendum, which is stipulated to be equal in meaning to the definiens. The definiens assigned contains nothing but old terms. In particular, the defined term is not used within the definiens.

Definitions of this kind are called *explicit definitions*.

Notion 1.42 (Explicit definition). An **explicit definition** is one in which the term being defined (in the definiendum) is not used within the definiens.

The restriction that the term being defined is not used within the definiens seems quite sensible, as we want to avoid that the following be regarded as a proper definition.

$$n \text{ is plok} \stackrel{\text{DF}}{=} n \text{ is plok} \quad (\approx)$$

The problem with $(\approx)$ is that it is useless as a definition. The meaning of the defined term 'plok' is not being set, because the definiens is not a meaningful expression at all.

Not all definitions must be explicit, though. There are cases in which it is useful to have the term defined appearing in the definiens. Consider, for instance, the following definition, corresponding to the mathematical definition of factorial[¶]:

¶: Note that, in the usual convention, 0 is not a natural numbers.

$$\begin{aligned} 0! &\stackrel{\text{DF}}{=} 1 && (\mathcal{D}_1) \\ (n)! &\stackrel{\text{DF}}{=} (n-1)! \times (n) && : n \text{ stands for a natural number.} && (\mathcal{D}_2) \end{aligned}$$

Not that, although in $(\mathcal{D}_2)$ we used the defined term '!', this does not generate a problem like $(\approx)$ does. The reason is that in $(\mathcal{D}_2)$ we are not merely repeating the term being defined, but providing a way to reduce each new case to cases previously defined.

First, the meaning of '0!' is fixed by $(\mathcal{D}_1)$. Second, by $(\mathcal{D}_2)$, the meaning of '1!' depends on the meaning of '(1-1)!', i.e., of '0!'. Next, the meaning of '2!' depends on that of '1!', and so forth. Thus, the meaning of each expression of the form 'n!', where n stands for 0 or a natural number, is determined by referring back to previous numerals, ultimately reaching the base case '0!'.

Thus, although the defined symbol occurs within the definiens of each instance of $(\succ_2)$, definition $(\succ)$ is problematic, because the meaning of each instance can be traced back to expressions whose meanings are already fixed. Definitions of these kinds are known as *recursive* or *inductive definitions*.

Notion 1.43 (Recursive definition). A **recursive** aka **inductive definition** stipulates the meaning of a term starting from **base cases** and then defining **inductive cases** depending on those base cases and the other inductive cases.

Thus, in the definition of factorial above, $(\succ_1)$ corresponds to the base case, while $(\succ_2)$ corresponds to the inductive case.

Historical note: Fibonacci numbers

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet:

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

Author (XXXX-XXXX)

Clause-based recursive definitions. Not all recursive definitions are easily stated in the form of (⋈). Suppose that we want to define the term ‘ancestor’ so that x is an ancestor of y in case x is a parent of y , or x is a grandparent of y , or x is a great-grandparent of y , or x is a great-great-grandparent of y , etc.

Here, we do not have one specific case from which to start. In definition (⋈), our base case was about ‘0’, but here, the base case consists is about all expressions expressing a parenthood relationship. Typically, we use a clause-based inductive definition, which may be described as follows.

Notion 1.44. A clause-based recursive definition consists of:

1. One or more *base clauses*, which introduce the simplest instances of the expression being defined.
2. One or more *inductive clauses*, which specifies how new instances arise from previously established ones.
3. A *final clause*, which states that all cases of the defined term are obtained from the base clauses with finitely many applications of the inductive clauses.

Let us see how a recursive definition may be given of ‘ancestor’.

Definition ȳ. Where ‘ x ’ and ‘ y ’ are variables standing for living beings, the term ‘ancestor’ is recursively defined by the following clauses:

1. x is a parent of $y \Rightarrow x$ is an ancestor of y . (*base clause*)
2. x is an ancestor of z and z is an ancestor of $y \Rightarrow x$ is an ancestor of y . (*inductive clause*)
3. No x is an ancestor of y unless this can be obtained by finitely many applications of 1 and 2. (*final clause*)

The base clause introduces the most immediate case of ancestry, namely that of parenthood. The inductive clause then extends the relation step by step: to x ’s parents, and to x ’s parents’ parents, and to x ’s parents’ parents’ parents¹. The final clause ensures that no additional cases are admitted as ancestors.

¹ And to x ’s parents’ parents’ parents’ parents.

Recursive definitions and definientia. Note that definition (ȳ) is not schematic in the sense of **notion 1.37**, since it contains no placeholders. As it contains variables, one might call it an open definition – and, thus, schematic in a sense.

Moreover, definition (ȳ) does not have the typical definiendum-definiens structure. Clearly, the expression ‘ x is an ancestor of y ’ is

the definiendum. However, where is its definiens? In order to see it more clearly, we might reformulate (¶) as follows:

$$\begin{aligned} \text{a is an ancestor of b} &\stackrel{\text{DF}}{=} \text{a is a parent of b} \vee \\ &\exists x (\text{a is an ancestor of } x \wedge \\ &\quad x \text{ is an ancestor of b}) \\ &: \text{a, b stand for living beings.} \quad (\text{¶}') \end{aligned}$$

Note that, unlike (¶), definition (¶') is schematic and has an easily identifiable definiens for each definitia of its instance definitions, all of which share the same form.

Moreover, note that, despite appearances, definition Equation ¶' is inductive, as its definiens states both the base case (to the left of '∨') and its inductive case (to its right).